which matches the pattern of Eq. 4.20 exactly.

> **Exercise 4.3:** Go back to the definition of Poisson brackets in *Volume I* and check that the identification in Eq. 4.21 is dimensionally consistent. Show that without the factor $\hbar$,it would not be.

Equation 4.21 solves a riddle. In classical physics, there is no difference between FG and GF. In other words: classically, commutators between ordinary observables are zero. From Eq. 4.21, we see that commutators in quantum mechanics are not zero, but that they are very small. The classical limit (the limit at which classical mechanics is accurate) is also the limit at which $\hbar$ is negligibly small. Therefore, it is also the limit at which commutators are very small in human units.

4.10 CONSERVATION OF ENERGY

How can we tell whether something is conserved in quantum mechanics? What do we even mean by saying that an observable—call it Q—is conserved? At the very minimum, we mean that its expectation value $\langle Q \rangle$ does not change with time (unless of course the system is disturbed). An even stronger condition is that $\langle Q^2 \rangle$ (or the expectation value of *any* power of Q) does not change with time.

Looking at Eq. 4.19, we can see that the condition for $\langle Q \rangle$ not to change is

$$[\mathbf{Q}, \mathbf{H}] = 0$$

In other words, if a quantity commutes with the Hamiltonian, its expectation value is conserved. We can make this statement stronger. Using the properties of commutators, it's easy to see that if $[\mathbf{H}, \mathbf{Q}] = 0$, then $[\mathbf{Q}^2, \mathbf{H}] = 0$, or even more generally, $[\mathbf{Q}^n, \mathbf{H}] = 0$, for any n. It turns out that we can make a stronger claim: if $\mathbf{Q}$ commutes with the Hamiltonian, the expectation